MODEL FITTING MNIST DATA
SUMMARY

BY JOHN STEELE

The MNIST (Modified National Institute of Standards and Technology) data was modeled by machine learning classifiers. The three models were Gauss Naive Bayes, Gauss Non-Naive Bayes, and KNN. The accucracy data scientist strive for is 87%. By only modifying the epsilon of the Gauss Non-Naive Bayes model, I was able to reach 95%.

- The following technologies were used: Python, Google Colab.
- The following Python libraries were used: NumPy, Pandas, MatPlotLib, and Seaborn.
- The MNIST data did not require any cleaning. Only needed to remove the first two columns because these were used for indexing and therefore not needed for the models.
- The highest accuracy obtained with Gauss Naive Bayes was 80.2%. This was with changing the epsilon. This maximum epsilon is 1000.
- The highest accuracy obtained with Gauss Non-Naive Bayes was 95.8%. This was with changing epsilon only. The maximum epsilon value is 1000.
- The KNN model did not complete as it took too long to run. I let it run for an hour and it did not complete so I stopped it. Since Non-Naive Bayes obtained 95.8%, KNN was not necessary.
- Future changes could include scaling the data and checking that all data is ‘correct’. ‘Correct’ being that the images match the labels.